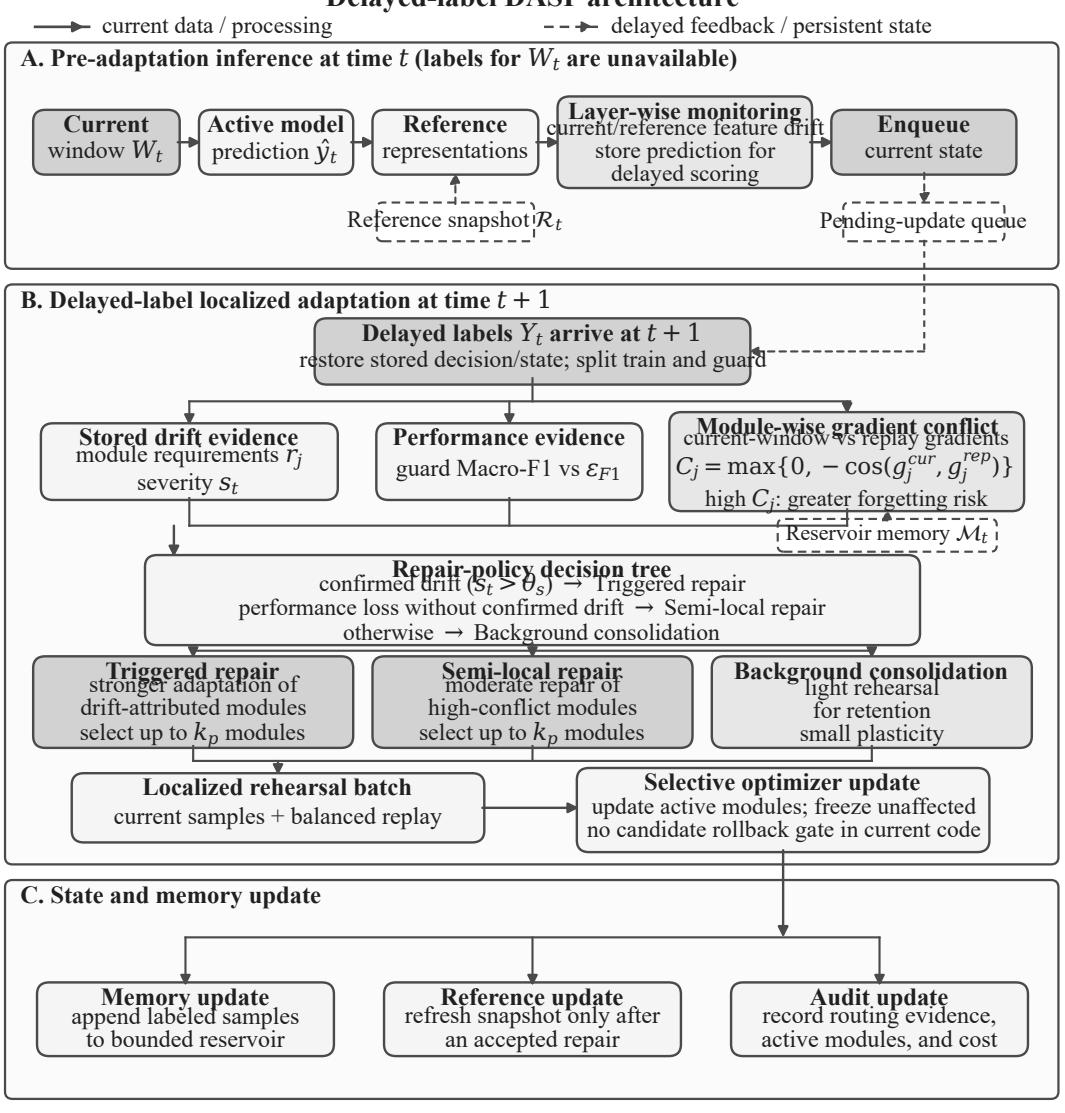

Delayed-label DASP architecture
current data / processing
delayed feedback / persistent state
A. Pre-adaptation inference at time t (labels for W_t are unavailable)
Current window W_t
Active model prediction $\hat{y}_t$
Reference representations
Layer-wise monitoring current/reference feature drift store prediction for delayed scoring
Enqueue current state
Reference snapshot $\mathcal{R}_t$
Pending-update queue
B. Delayed-label localized adaptation at time $t+1$
Delayed labels Y_t arrive at $t+1$ restore stored decision/state; split train and guard
Stored drift evidence module requirements r_j severity s_t
Performance evidence guard Macro-F1 vs ε_{F1}
Module-wise gradient conflict current-window vs replay gradients $C_j = \max\{0, -\cos(g_j^{cur}, g_j^{rep})\}$ high C_j: greater forgetting risk
Reservoir memory $\mathcal{M}_t$
Repair-policy decision tree confirmed drift ($s_t > \theta_s$) → Triggered repair performance loss without confirmed drift → Semi-local repair otherwise → Background consolidation
Triggered repair stronger adaptation of drift-attributed modules select up to k_p modules
Semi-local repair moderate repair of high-conflict modules select up to k_p modules
Background consolidation light rehearsal for retention small plasticity
Localized rehearsal batch current samples + balanced replay
Selective optimizer update update active modules; freeze unaffected no candidate rollback gate in current code
C. State and memory update
Memory update append labeled samples to bounded reservoir
Reference update refresh snapshot only after an accepted repair
Audit update record routing evidence, active modules, and cost